C# BACKEND TRACK | OOP PHASE

Task 1

System Analysis Task

Think about the business. Extract the models. Assign the sources.

Codeline Bootcamp · Backend Phase · OOP Session 3 Task

SOURCE TAG LEGEND

Every property of every class model gets its value from somewhere. Before writing any code, you must decide **where** each value comes from. Use one of these five source tags to label each property in your analysis:

System Generated	**System Generated**	The system assigns this value automatically. The user never types it. Example: an auto-incremented ID calculated as List.Count + 1.
User Input	**User Input**	The user types this value directly at the keyboard when prompted by the system.
Default Value	**Default Value**	A fixed value set by the code when the object is created. The user never chooses it. Example: isBooked = false, status = "Active".
Calculated	**Calculated**	Derived automatically from another object already in memory — no user prompt. Example: appointment date copied from a chosen slot, fee copied from a doctor.
From List	**From List**	The user selects a value from a list of options displayed on screen. Example: choosing a doctor ID from a printed list of doctors.

Domain 1 — Library Management System

Extract everything yourself

Read the following description of a real library carefully. Your job is to go through the 6-phase process and fill in all the blank tables below. Nothing has been done for you — you identify the entities, you decide the properties, and you assign the source tag for each one.

> A public library allows community members to register and borrow books. Each book has a title, author, genre, and a fixed number of physical copies in stock. A registered member can borrow an available book copy and must return it within 14 days. If the book is returned late, the system calculates a fine based on how many days overdue it is. A member can also reserve a book that is currently fully borrowed so they are notified when it becomes available. The librarian needs to see all currently active loans, view all overdue loans sorted by how late they are, and get a report of the most-borrowed books ranked by total borrow count.

Step 1 — Identify the Entities (Phases 1 & 2)

From the description above, identify the main real-world objects (classes) the system must track. Write each entity name, its properties, the data type for each, and the source tag that describes how each property gets its value.

Entity Name: Book

Property Name	Data Type	Source Tag
BookID	int	System Generated (auto)
BookTitle	string	user input
BookAuthor	string	user input
BookGener	string	user input
Total_Copies	int	user input
AvailableCopies	int	Calculated
is_available	bool	Calculated

Entity Name: Member

Property Name	Data Type	Source Tag
MemberID	int	System Generated (auto)
MemberName	string	user input
MemberEmail	string	user input
RegisteredDate	Date	System Generated

Entity Name: Loan

It mean system store Borrow Record

Property Name	**Data Type**	**Source Tag**
LoanID	int	System Generated (auto) every class/entity need auto genereation ID
BorrowDate	Date	System Generated Auto will store the today date for borrow
DueDate	Date	Calculated BorrowDate +14 days loan.BorrowDate.AddDays(14);
ReturnDate	Date	System Generated set automatically when the member returns the book , if not return it will be empty
OverDue_date	int	Calculated I use (int) because it will represent whole number of days not date
fine	Decimal	Calculated Fine = OverDue_date× FinePerDay.
memberID	int	from list
BookID	int	from list

Entity Name: Reservation

Property Name	**Data Type**	**Source Tag**
ReservationID	int	System Generated (auto)
ReservationDate	Date	System Generated
Notified	bool	Calculated If available will notified
MemberID	int	from list
BookID	int	from list

If you identify more than 4 entities, continue on the back of this sheet or add extra rows above.

Step 2 — Describe the Entity Relationships (Phase 1 continued)

How do the entities connect to each other? Describe the full cycle in plain English — who does what, which entity gets created, and what changes in other entities as a result.

Write the relationship cycle here:

A Member registers in the library and can borrow or reserve a Book, so the system creates a Loan or Reservation, updates book availability, records return date, calculates fine if late, and notifies the member when the book becomes available.

Step 3 — List the Services (Phase 4)

From the description, identify every service (function) the system needs. Write at least 8 services — give each a name and a one-line description of what it does.

#	Service Name	What it does
1	Register Member	Adds a new member in library
2	Add Book	Adds a new book with needed information
3	Borrow Book	Creates a loan when a member borrows a book.
4	Reserve Book	Creates a reservation if book is not available and notified when be available
5	View Member Details	Shows information about the member.
6	View Book Details	Shows information about one book and its availability.
7	View Loans	Shows all books currently borrowed and not returned
8	Most Borrowed Books Report	Shows books ranked by total borrow count.
9	Search Book	Finds a book by title or author
10	Update Member Information	Changes member details. ex: email or name

Domain 2 — Hotel Management System

Assign the source tags

The class models and all their properties are already defined for you below. Your job is to **fill in the Source Tag column** for every single property. Use the five source tags from the legend on page 1: System Generated, User Input, Default Value, Calculated, From List.

A hotel allows guests to register and book rooms for a stay. The hotel has different room types (Single, Double, Suite) each with its own nightly price. Each room can only have one active booking at a time. When a guest checks in, a check-in record is created. The total price of a stay is calculated from the number of nights and the room's nightly rate. When a guest checks out, the room becomes available again. A booking can also be cancelled before check-in, which frees the room immediately.

Your task: For each property in the tables below, write the correct source tag in the empty column on the right. Think carefully — ask yourself: does the user type this? does the system generate it? is it copied from another object? does it have a fixed starting value? does the user pick it from a list?

Guest

Property Name	Data Type	Source Tag (fill in)
`guestId`	int	System Generated
`guestName`	string	User Input
`guestPhone`	string	User Input
`guestEmail`	string	User Input
`guestNationality`	string	User Input
`registrationDate`	string	System Generated

Room

Property Name	Data Type	Source Tag (fill in)
`roomId`	`int`	System Generated
`roomNumber`	`string`	User Input
`roomType`	`string`	User Input
`floorNumber`	`int`	User Input
`nightlyPrice`	`decimal`	User Input
`isAvailable`	`bool`	Calculated

Booking

Property Name	Data Type	Source Tag (fill in)
`bookingId`	`int`	System Generated
`guestId`	`int`	from list
`roomId`	`int`	from list
`checkInDate`	`string`	System Generated
`checkOutDate`	`string`	Calculated
`numberOfNights`	`int`	Calculated
`totalPrice`	`decimal`	Calculated
`status`	`string`	Calculated

CheckInRecord		
Property Name	**Data Type**	**Source Tag (fill in)**
`recordId`	int	System Generated
`bookingId`	int	from list
`guestId`	int	from list
`roomId`	int	from list
`actualCheckIn`	string	System Generated
`actualCheckOut`	string	System Generated
`finalAmount`	decimal	Calculated

Step 2 — Describe the Entity Relationships

Now that you have studied the four class models, describe in your own words how they connect to each other. Who creates what? What changes when a booking is made? What happens when a guest checks out?

Write the relationship cycle here:

A guest registers and books a room, so a Booking is created, the room becomes unavailable, then check-in creates a CheckInRecord, and check-out calculates the final amount and makes the room available again.

Step 3 — List the Services

Based on the description and the class models above, list all the services (functions) this hotel system needs. Give each a name and a one-line description.

#	Service Name	What it does
1	Register Guest	Adds a new guest
2	Add Room	Adds a new room with type and price.
3	View Available Rooms	Shows rooms that not booked yet
4	Book Room	Creates a booking .
5	Check In Guest	Creates a check-in record when guest arrives.
6	Check Out Guest	Records checkout, calculates final amount, and makes room available.
7	Cancel Booking	Cancels booking before check-in and frees the room.

#	Service Name	What it does
8	Update Guest Information	Updates guest details like phone or email.
9	Booking Report	Shows all bookings with guest, room, dates, status, and total price.